MILITARY SPECIFICATION SHEET

INTERCOMMUNICATION SETS AN/AIC-14 AND AN/AIC-14A

PASSIVE INTERPHONE CONTROL PANEL

C-2644/AIC-14

Inactive for new design after
For new design use MIL-I-22352/17(AS) or MIL-I-22352/18(AS).

This specification sheet is approved for use by the Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the unit described herein shall consist of this document and the latest issue of specification MIL-I-22352(AS).

1. SCOPE

This specification sheet provides performance and physical characteristics for the passive Interphone Control panel C-2644/AIC-14, hereinafter referred to as C-2644.

1.2 Military Part Number. The military part number shall be M22352/12.

2. APPLICABLE DOCUMENTS

Refer to 20 of MIL-I-22352(AS) Appendix A for a listing of applicable documents. The requirements specified herein shall have precedence over all referenced documents. The contract and its attachments shall have precedence over any specification or reference document.

3. REQUIREMENTS

3.1 Item Definition. The passive interphone control panel shall provide the operator with the capability to activate interconnections with on-board receivers and transmitters. The operator using the C-2644 shall have the capability to interact with the associated radio transmitters and receivers and to have access to both interphone signals. The operator shall have the capability of selecting from the OFF mode, a receive mode

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from five associated receivers, for keying and connecting of transmit audio to one selected transmitter of the four available. Interphone volume control shall also be provided.

3.1.1 Item Diagram. Passive Control Panel C-2644 shall provide the functions represented schematically in accordance with the diagram of Figure 1.

3.1.2 Interface definition. The passive Interphone Control panel input and output signals will normally exhibit the characteristics presented in Table I.

TABLE I. Interface signal list.

Signal Name	Characteristics
Inputs:	
Receiver No. 1 (RCVR NO. 1)	Voice range audio (300 to 6,000 Hz) signal ranging in amplitude from 0 to 15 Vac. No dc component normally present.
Receiver No. 2 through Receiver No. 5 (RCVR NO. X)	Same as Receiver No. 1
Interphone No. 1 (INTPH NO. 1)	Voice range audio (300 to 6,000 Hz) signal ranging in amplitude from 0 to 1 Vac. No dc component normally present.
Interphone No. 2 through Interphone No. 5 (INTPH NO. X)	Same as Interphone No. 1
Transmit Audio (XMIT AUDIO)	Voice range audio (300 to 6,000 Hz) signal ranging in amplitude from 0 to 1 Vac.

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TABLE I. Interface signal list - continued.

Signal Name	Characteristics
Transmit Key (XMIT KEY)	Relay logic and current sink for lamps having two states: 1) Open and 2) Short to ground
Panel Lights (LIGHTS)	0 to + 28Vdc or 0 to 6V, 400 Hz
Outputs:	
Interphone (INTPH)	Voice range audio (300 to 6,000 Hz) signal ranging in amplitude from 0 to 1 Vac. No dc component normally present.
Selected Receiver (SEL RCVR)	Voice range audio (300 to 6,000 Hz) signal ranging in amplitude from 0 to 1 Vac. No dc component normally present.
Transmit Audio No. 1 (XMIT AUDIO NO. 1)	Voice range audio (300 to 6,000 Hz) signal ranging in amplitude from 0 to 1 Vac. No dc component normally present.
Transmit Audio No. 2 through Transmit Audio No. 4 (XMIT AUDIO NO. X)	Same as (XMIT) Audio No. 1
Transmit Key No. 1 (XMIT KEY NO. 1)	Relay logic: 1) lamp supply voltage 2) short to ground
Transmit Key No. 2 through Transmit Key No. 4 (XMIT KEY NO. X)	Same as Transmit Key No. 1
Emergency Receiver No. 1 (EMER RCVR NO. 1)	Voice range audio (300 to 6,000 Hz) signal ranging in amplitude from 0 to 15 Vac. No dc component normally present.
Emergency Receiver No. 2 and No. 3 (EMER RCVR NO. X)	Same as Emergency Receiver No. 1

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TABLE I. Interface signal list - continued.

Signal Name	Characteristics
Grounds:	
Power (PWR GND)	Power return. See 3.4.4.2 of MIL-I-22352(AS).
System (SYS GND)	Signal return. See 3.4.4.3 of MIL-I-22352(AS).

3.1.3 Controls. The passive Interphone Control panel shall provide the controls listed in Table II.

TABLE II. C-2644/AIC-14 controls list.

Title	Function
TRANS SEL	The C-2644 shall provide a four-position rotary switch for the selection of one of four transmitters for keying and connection of the transmit audio.
RCVR SEL	The C-2644 shall provide five two-position toggle switches with the following functions: OFF - In the down position the associated receiver output is disabled. ON - In the up position the system shall present the audio signals from the associated receiver.
RAD VOL	The C-2644 shall provide a control for the continuous adjustment of radio signal output level.

3.1.4 Panel Lamps. The passive Interphone Control panel shall provide the three (edge lighted) panel lamps to illuminate the front panel.

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3.2 Characteristics

3.2.1 Performance Characteristics

3.2.1.1 Input Characteristics

3.2.1.1.1 Receiver No. 1 through No. 5 (RCVR NO. X). The signals incoming from radio receivers shall be identified and assigned as a function of the aircraft in which the system is installed. The RCVR input characteristics shall be as follows:

3.2.1.1.1.1 Signal Frequency. The RCVR NO. X inputs shall accept drive signals which range from 300 to 6,000 Hz.

3.2.1.1.1.2 Signal Amplitude. The RCVR NO. X inputs shall accept drive signals which range in amplitude from 0 to 15 Vac.

3.2.1.1.1.3 Input Impedance. The RCVR NO. X inputs shall each present a 10,000 ohm ±10 percent impedance to the incoming signals when the ON (up) position is selected on the RCVR SEL switch otherwise open.

3.2.1.1.1.4 Isolation. The isolation between RCVR NO. X signals shall be at least 68 dB for signals ranging from 300 to 6,000 Hz.

3.2.1.1.2 Interphone No. 1 through Interphone No. 5 (INTPH NO. X). The INTPH NO. 1 through INTPH NO. 5 input characteristics shall be as follows:

3.2.1.1.2.1 Signal Frequency. The INTPH NO. X input shall accept drive signals which range from 300 to 6,000 Hz.

3.2.1.1.2.2 Signal Amplitude. The INTPH inputs shall accept drive signals which range in amplitude from 0 to 10 Vac.

3.2.1.1.2.3 Input Impedance. The INTPH inputs shall each present a 62,000 ohm ±10 percent impedance to the incoming signals.

3.2.1.1.2.4 Isolation. The isolation between RCVR signals shall be at least 68 dB for signal ranging from 300 to 6,000 Hz.

3.2.1.1.3 Transmit Audio (XMIT AUDIO). The XMIT AUDIO signal shall be applied to the TRANS SEL switches. The XMIT AUDIO input shall exhibit the following characteristics:

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3.2.1.1.3.1. Signal Frequency. The XMIT AUDIO input shall accept drive signals which range from 300 to 6,000 Hz.

3.2.1.1.3.2. Signal Amplitude. The XMIT AUDIO input shall accept drive signals which range in amplitude from 0 to 10 Vac.

3.2.1.1.4 Transmit Key (XMIT KEY). The XMIT KEY input shall accept a relay drive logic signal having two states: 1) open and 2) short to ground. The XMIT KEY signal shall be used to produce the XMIT KEY NO. 1 through NO. 4 output on the line associated with the transmitter selected on the TRANS SEL switch.

3.2.1.1.5 Panel Lights (LIGHTS). The C-2644 shall be capable of accepting power ranging up to +28 Vdc or 6 Vac to illuminate the three front panel lamps. The panel lamps used shall be selected accordingly from TABLE III.

TABLE III. Front panel bulbs.

Nomenclature	Power
MS25237A-327	+28 Vdc
MS25237A-328	6 Vac

3.2.1.2 Output Characteristics. All signals which are passed through this unit from input to output by being routed only through connectors, wire, printed wiring, switches, and relays shall not experience insertion loss in excess of 0.5 dB. The C-2644 passive Interphone Control Panel shall have the following output characteristics.

3.2.1.2.1 Interphone (INTPH). The INTPH output shall be the sum of INTPH NO. 1 through NO. 5 signals each reduced by 40.4 ± 1dB in amplitude from the individual input value(s) and summed when terminated in 600 ohms. The contribution to the total by each input shall be equal ± 10 percent.

3.2.1.2.2 Selected Receiver (SEL RCVR). The SEL RCVR output shall be the sum of the operator selected RCVR NO. 1 through RCVR NO. 5 inputs adjusted (reduced) in amplitude by the RAD VOL control. Each RCVR input shall contribute equally ± 10 percent to the total output. The RAD VOL control shall demonstrate an attenuation range of 30 dB.

3.2.1.2.3 Transmit Audio No. 1 through Transmit Audio No. 4 (XMIT AUDIO NO. X). Only one of the XMIT AUDIO NO. 1 through XMIT AUDIO NO. 5 outputs shall have an active signal at any one time. The XMIT AUDIO output line shall be taken from the XMIT AUDIO input. The specific active line to present the output shall be determined by operator selection of a RCVR SEL switch to the ON position, the leftmost switch having priority.

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3.2.1.2.4 Transmit Key NO. 1 through Transmit Key NO. 4 (XMIT KEY NO. X). The XMIT KEY No. 1 output shall be a relay drive logic signal having two states: 1) lamp supply voltage and 2) short to ground. The output shall be a short to ground when the associated RCVR SEL switch is the leftmost switch selected to ON position and the XMIT KEY input is a ground. XMIT KEY NO. 2 through XMIT KEY NO. 5 signal shall exhibit equivalent characteristics as XMIT KEY NO. 1.

3.2.1.2.5 Emergency Receiver No. 1 and Emergency Receiver No. 2 and Emergency Receiver No. 3 (EMER RCVR NO. X). The EMER RCVR NO. X output shall be connected to the RCVR NO. X input through the associated RCVR SEL switch withih the C-2644. Therefore, the output characteristics shall be identical to the input.

3.2.2 Physical characteristics.

3.2.2.1 Dimensions. The C-2644 shall consist of a single panel-mounted assembly having the dimensions of Figure 2. All outside dimensions shall be within ±0.03 inches of the nominal value shown.

3.2.2.2 Weight. The C-2644 shall weigh no more than 2.0 pounds.

3.3 DESIGN AND CONSTRUCTION

The design and construction of the passive Interphone Control panel C-2644 shall be in accordance with the requirements of 3.3 of MIL-I-22352(AS).

3.3.1 Front panel lighting. The front panel lamps shall be installed using mounting hardware having red filters in accordance with MIL-L-25467 to enhance night vision. The panel lamps shall be mounted in accordance with the outline drawing Figure 2. The lamp drive circuitry shall be capable of accommodating power ranging up to 28 Vdc or 6 V, 400 Hz ac.

4. QUALITY ASSURANCE PROVISIONS

See MIL-I-22352(AS).

4.1 Responsibility for tests and inspections. See MIL-I-22352(AS).

4.2 Test plan. A detailed test plan will be submitted by the contractor for approval by the acquiring activity which details the tests to be performed in order to verify compliance with the performance characteristics specified in 3.2.1. The test plan will consider all of the inspections delineated in 4.2 of MIL-I-22352(AS).

4.3 Test procedures. Detailed test procedures will be submitted by the contractor for approval by the acquiring activity which will satisfy the requirements of the test plan of 4.2.

Preparing Activity:
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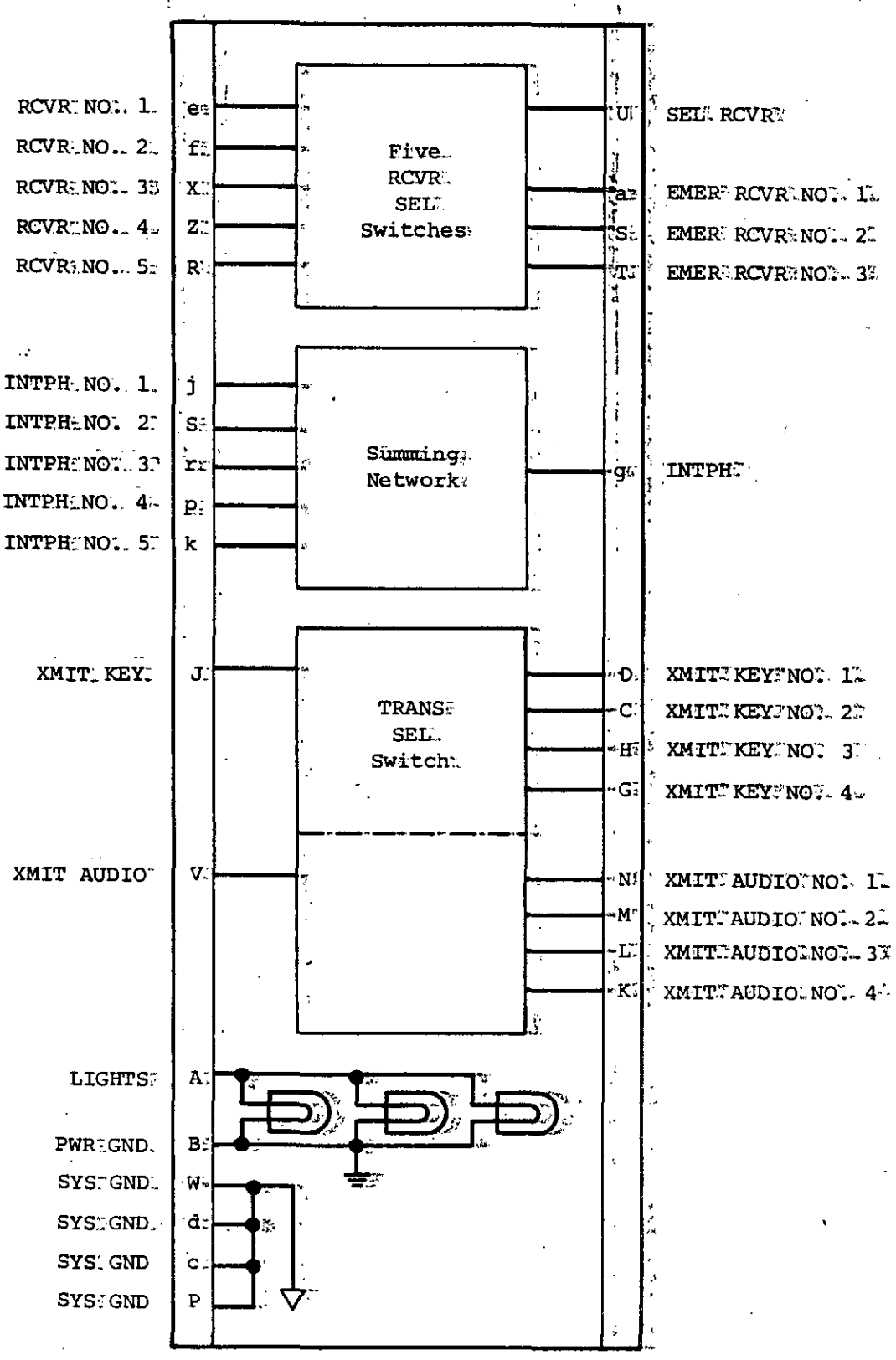
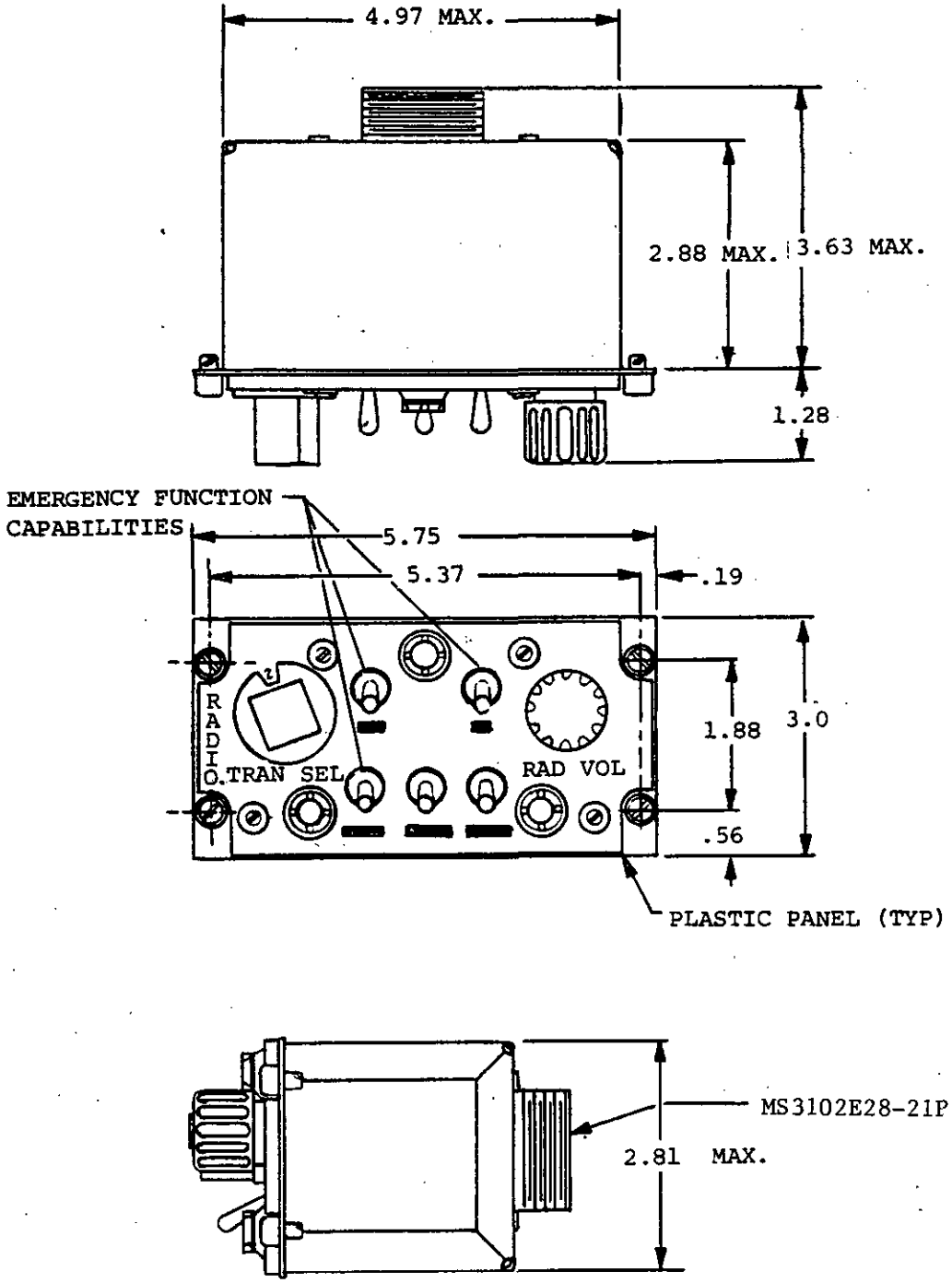


FIGURE 1. C-2644 functional diagram.

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NOTE:

1 All dimensions are in inches. Tolerances on dimensions are ±0.03 inches unless otherwise specified.

FIGURE 2. C-2644 outline drawing.

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1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-I-22352/12(AS)	INTERCOMMUNICATION SETS AN/AIC-14 AND AN/AIC-14A PASSIVE INTERPHONE CONTROL PANEL C-2644/AIC-14

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
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